

# Horizon Quantum Gravity (HQGG) Cosmology: Space, Frequency, Spheres, and the Nature of Dark Matter and Dark Energy

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## Abstract

We present a complete cosmological model within the HQGG framework, in which space-time is treated as an elastic medium (fabric) with local frequency. Gravity is a gradient of this frequency, time is inverse frequency, and matter is standing waves. The Universe originates from the Void-Home  $S$  through implosive quantization and sequential unfolding of 10 quantum spheres. Our sphere  $S^1$  contains 10 internal modes, structured by the 3+7 law. Dark matter is explained as local enhancement of waves near massive objects, and dark energy as global frequency attenuation. The model yields quantitative predictions for the CMB, large-scale structure, and gravitational waves, without free parameters.

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# 1 Introduction

The standard  $\Lambda$ CDM model relies on two dark entities — dark matter and dark energy — whose nature remains unknown. Moreover, it leaves unresolved the singularity problem, the origin of structure, and the anomalies in the CMB spectrum.

In this work we propose an alternative: space-time is a medium with a local frequency. This approach, developed within the Horizon Quantum Gravity Group (HQGG), unifies gravity, quantum physics, and cosmology.

## 2 The Fabric of Space

### 2.1 Definition

Space-time is an elastic medium — the fabric — characterized by a local frequency  $\omega(x^\mu)$ . This frequency determines all physical properties.

### 2.2 Gravity as Frequency Gradient

$$\mathbf{g} = \frac{1}{\omega_0} \nabla \omega$$

Particles move toward higher frequency, i.e., toward denser fabric.

### 2.3 Field Equation

$$\partial_\mu \partial_\nu \omega = \frac{8\pi G}{c^4} T_{\mu\nu} \cdot \omega$$

### 2.4 Metric from Frequency

$$g_{\mu\nu} = \eta_{\mu\nu} + \frac{1}{\omega_0} \partial_\mu \partial_\nu \omega$$

## 2.5 Time as Inverse Frequency

$$t = \frac{1}{\omega}$$

Time flows faster near massive objects; the observed slowing is an illusion.

## 3 Cosmological Hierarchy: From the Void to the Spheres

### 3.1 The Void-Home $S$

The singularity is replaced by the Point of Creation and the Void-Home, formed by spherical compression of a tiny part of infinite space. The Void has no modulations; it is pure potentiality.

### 3.2 Implosive Wave and 10 Quantum Spheres

An implosive wave quantizes the Void into 10 discrete modes — the quantum spheres:

$$S \rightarrow S^1 \rightarrow S^2 \rightarrow \dots \rightarrow S^{10}$$

$S^1$  is our Universe;  $S^{10}$  is the outer boundary.

### 3.3 Internal Modes of $S^1$

Inside  $S^1$ , 10 internal modes  $S_1^1 \dots S_{10}^1$  are defined by:

$$\omega_n = \omega_1 \cdot \frac{11-n}{10}, \quad n = 1, \dots, 10$$

We are currently at  $S_5^1$ , with 60% frequency drop and 22% of time elapsed.

### 3.4 The 3+7 Law

Three frequency states:

- 0 — base (Planck level),
- +1 — trans-Planckian (compressed fabric),
- -1 — sub-Planckian (relaxed fabric).

Their combinations yield 7 stable manifestations: colors, notes, shells, ranges, days, etc.

## 4 Cosmological Evolution and Dark Entities

### 4.1 Frequency Drop and Expansion

$$\begin{aligned}\omega(t) &= \omega_1(0) \cdot e^{-\gamma t} \\ a(t) &= a_0 \cdot \exp\left(\frac{\gamma}{2}t^2\right)\end{aligned}$$

### 4.2 Dark Matter

Dark matter is local wave enhancement near massive objects:

$$\rho_{\text{DM}} = \rho_0 \left(1 + \frac{\Phi}{c^2}\right)$$

### 4.3 Dark Energy

Dark energy is global frequency attenuation:

$$\dot{\omega} = -\gamma\omega$$

## 5 Observational Predictions

- CMB: suppression of  $C_2, C_3$ , axis of evil, hemispherical asymmetry.
- Quasars: peak at  $z \sim 2\text{--}3$ , disappearance at  $\sim 3$  billion light-years.
- Gravitational waves: modulation  $h(t) = h_{\text{chirp}}(t) \cdot (1 + \epsilon \cos(\Delta\omega t))$ .
- Large-scale structure:  $\xi_{\text{galaxies}} \propto \omega^{-3}$ .

## 6 Conclusion

HQGG cosmology provides a complete, self-consistent description without dark entities, relying only on  $G, \hbar, c$ .

**Space is not curved — it has frequency.**  
**Time is not absolute — it is inverse frequency.**  
**Matter is waves, gravity is their gradient.**  
**This is HQGG.**

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## References

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